MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, M712
GUIDANCE SECTION FOR

This amendment forms a part of Military Specification MIL-P-63224A (AR) dated 14 May 1986, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Government documents. Add the following Military Specification:

"MIL-H-6875 - Heat Treatment of Steels (Aerospace Practice, Process For)".

PAGE 2

Add paragraph 2.2 as follows:

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply;

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8 - Tension Testing of Metallic Materials
E18 - Tests for Rockwell Hardness of Metallic Materials
A370 - Standard Methods and Definitions for Mechanical Testing of Steel Products

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ANSI-B46.1-1962 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)"

AMSC N/A FSC 1320

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Add paragraph 3.6 as follows:

"3.6 Electronic housing integrity.

3.6.1 Fluorescent penetrant examination. There shall be no indications of cracks, laps, seams, or tears when the fully machined part is inspected by fluorescent penetrant inspection.

3.6.2 Mechanical properties. Mechanical properties of the electronic housing shall be as required on the applicable drawing."

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Add paragraph 4.4.1.1 as follows:

"4.4.1.1 Heat treatment lot formation. Heat treatment lots shall be defined as electronic housings heat treated in one continuous run (see 4.4.1.1.2), and shall not exceed one mill heat of steel, 600 components, or 24 hours production, whichever is smallest. Additionally, heat treatment lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time, and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into large heat treatment lots after qualifying with individual batches, provided qualifying procedures are submitted to and approved by the Government and provided all other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment lots shall remain grouped together and identified by lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots. The contractor may submit an alternate procedure for heat treatment lot formation and inspection of the heat treatment lot to the procuring activity for approval.

4.4.1.1.2 Continuous run. A continuous run is defined to be a run which involves only one part (i.e. control housing, warhead body or electronic housing), with no interruption of the scheduled flow of parts into and out of the system and no changes in the parameters of any furnace or bath."

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PAGES 43 and 44

4.4.2.17 Delete and substitute new enclosed paragraph (Sheets 1 and 2)

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Add paragraph 4.4.3.2 as follows:

"4.4.3.2 Mechanical properties of electronic housing.

4.4.3.2.1 Hardness. Each electronic housing shall be Rockwell hardness tested per paragraph 4.5.5.1. These values will also be used for selection of tension test samples of 4.4.3.2.2.

4.4.3.2.2 Tensile test. Two electronic housings from each heat treatment lot shall be selected for this test (one body with the highest hardness reading and one with the lowest hardness reading). From each of these bodies specimens shall be tested per paragraph 4.5.5.2 to determine compliance with tensile strength, yield strength and elongation requirements. Failure of any tensile specimens shall be cause for rejection of the heat treat lot.

4.4.3.2.3 Retest. If the high hardness body fails the yield strength or minimum ultimate strength requirement, the entire heat treatment inspection lot shall be rejected. In the event the lowest hardness body fails to meet the yield strength or minimum ultimate strength requirement, higher hardness values may be selected and tested in accordance with 4.4.3.2.2 in order to establish the acceptable minimum hardness. Once this minimum hardness has been established, all bodies with hardness values below the new minimum shall be rejected. If the low hardness body fails the elongation requirement, the entire heat treatment inspection lot shall be rejected. In the event the highest hardness body fails to meet the elongation requirement, lower hardness values may be selected and tested in accordance with 4.4.3.2.2 in order to establish the acceptable maximum hardness. Once this maximum hardness has been established, all bodies with hardness values above the new maximum shall be rejected."

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Add paragraph 4.5.5 as follows:

"4.5.5 Electronic housing.

4.5.5.1 Hardness. In order to determine uniformity of heat treatment, each housing after final heat treatment shall be subjected to a hardness test with a Rockwell Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the rear face of the electronics housing. The test shall be made using the procedures for performing this test as prescribed by ASTM-E18 and ASTM-A370.

4.5.5.2 Tensile strength, yield strength, and percent elongation. The hardest and softest electronic housings as determined by the hardness test in 4.5.5.1 from each heat treatment lot shall be subjected to this test. Four (4) test specimens shall be selected from each housing. The specimens shall be selected approximately 90° apart and shall be the largest obtainable longitudinal specimen as prescribed by ASTM E-8. The number of specimens and their location are for the convenience of the contractor and the Government and shall not restrict the contractor or Government from testing specimens at other locations to determine that the mechanical properties of the housings meet the drawing requirement. The criteria for performing the tests shall be as prescribed by ASTM E-8.

4.5.5.2.1 Elongation. Acceptance shall be based on the average elongation of all tensile specimens selected from each housing. If the average elongation is less than 10 percent, then the lot represented by this test sample shall be rejected.

4.5.5.3. Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be performed on 100 percent of the housings after final machining and heat treating. Inspection shall be in accordance with Group VI, MIL-I-25135 and MIL-I-6866. Both sides of the part shall be examined. Parts examined by penetrant inspection and found acceptable shall be ink stamped or stenciled with the letter "P". The marking shall withstand normal handling."

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1320-A979)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9306520
4.4.2.17	HOUSING, ELECTRONICS				NEXT HIGHER ASSEMBLY 9301790
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bourrelet diameter		0.40%	Dwg.	Gage
102	Hardness		100%	Dwg.	4.5.5.1
103	Mechanical properties		4.4.3.2.2	3.6.2	4.5.5.2
104	Penetrant inspection		100%	3.6.1	4.5.5.3
Minor					
201	Runout between I.D. and bourrelet .O.D		0.65%	Dwg.	Gage
202	Positional tolerance of aft end O.D. with respect to inside diameter of housing		0.65%	Dwg.	Gage
203	Positional tolerance of aft end I.D. with respect to inside diameter of housing		0.65%	Dwg.	Gage
204	Inside diameter of housing		0.65%	Dwg.	Gage
205	Positional tolerance of forward end I.D. with respect to inside diameter housing		0.65%	Dwg.	Gage
206	Runout between forward housing I.D. and aft end splice O.D.		0.65%	Dwg.	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9306520
4.4.2.17	HOUSING, ELECTRONICS				NEXT HIGHER ASSEMBLY 9301790
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Runout between forward threads and housing I.D.		0.65%	Dwg.	Gage
208	Positional tolerance of forward splice holes with reference to forward I.D., and forward edge of locating slot.		0.65%	Dwg.	Gage
209	Positional tolerance of aft splice holes with reference to aft splice O.D. and aft splice shoulder		0.65%	Dwg.	Gage
210	Parallelism of aft vertical splice surface with respect to forward end		0.65%	Dwg.	Gage
211	Dye penetrant inspection		100%	Dwg.	MIL-I-6866
212	Workmanship		0.65%	3.5	Visual

NOTES:

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